Computer Science - Undergraduate Level Assessment 2 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Which of the following statements is FALSE about memory reclamation based on reference counting?
   (a) Reference counting is well suited for reclaiming cyclic structures.
   (b) Reference counting incurs additional space overhead for each memory cell.
   (c) Reference counting is an alternative to mark-and-sweep garbage collection.
   (d) Reference counting need not keep track of which cells point to other cells.

2. Of the following, which gives the best upper bound for the value of f(N) where f is a solution to the recurrence f(2 N + 1) = f(2 N) = f(N) + log N for N >= 1, with f(1) = 0?
   (a) O(log N)
   (b) O(N log N)
   (c) O(log N) + 0(1)
   (d) O((log $N)^2$)

3. In Python 3, which of the following function sets the integer starting value used in generating random numbers?
   (a) choice(seq)
   (b) randrange ([start,] stop [,step])
   (c) random()
   (d) seed([x])

4. Which of the following is not a sentence that is generated by the grammar A -> BC, B -> x|Bx, C -> B|D, D -> y|Ey, E -> z?
   (a) xyz
   (b) xy
   (c) xxzy
   (d) xxxxy

5. Let x = 8. What is x>>1 in Python 3?
   (a) 3
   (b) 4
   (c) 2
   (d) 8

True / False (10 marks)

Answer True or False

6. True or False: The function O($N^{(}$1/N)) grows the slowest among the given function types as N increases.

7. True or False: The boolean expression a[i] == max || !(max != a[i]) simplifies to a[i] != max, indicating inequality with max.

8. True or False: Confidentiality is breached when someone logs into a system with a stolen username and password.

9. True or False: An Internet Protocol (IP) address is assigned to a device when it connects to the Internet.

10. True or False: The presence of several different classes of registers is not generally an obstacle to aggressive pipelining of an integer unit.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function that matches a string that has an a followed by zero or more b's by using regex.

12. Write a function to find the surface area of a cube.

End of Paper